

There are a lot of subfields in retail that AI has been a part of. These subfields are driven through the needs of consumers, businesses, and technology as well.

These key drivers can range from:

Consumer Expectations

- Personalization
- Convenience
- Speed

Business Pressures

- Competition
- Margins
- Efficiency

Consumer Expectations

- Computing Power
- Data Availability

To review, some of the core AI technologies that are also used in retail are machine learning, computer vision, natural language processing, robotics, and predictive analytics.

Computer Vision can be applied in retail in the form of shelf monitoring, customer tracking, self-checkout systems, loss prevention, and visual searches.

As for Natural Language Processing, that can be applied in retail in the form of customer service chatbots, voice shopping, social media monitoring, customer feedback analysis, and product search optimizations.

Virtual and Augmented Reality Technology can be applied in retail in the form of virtual try-on, store navigation, product visualization, interactive displays, and training applications.

There are also some Smart Store Technologies as well, like smart shelves, interactive mirrors, electronic shelf labels, automated checkout, traffic analysis

Supply Chain AI also has been used in the form of demand forecasting, inventory optimization, route optimization, supplier management, and risk prediction.

There are also some technologies in Warehouse Automation such as picking robots, Automated Guided Vehicles (or AGVs), sorting systems, inventory drones, and smart storage.

AI has also been used for store operations such as staff scheduling, maintenance prediction, energy optimization, security systems, and cleaning automation.

Though all of this is not without its challenges. From data quality issues, to integration difficulties, to change in management, ROI measurements, and skill gaps, there are still many hurdles that need to be jumped in order for AI to smoothly be integrated into retail.

There are also always going to be some ethical concerns that need to be addressed, like privacy, bias, employment impact, data security, and transparency.

However, while those challenges and concerns are going to be something we need to tackle, the current progress of AI in retail shows that this path may not be just a thought experiment.